

Northbridge Institute of Advanced Computing

AI Research Handbook

Edition 4.2 - Effective Academic Year 2025-2026

Issued by the Office of Research Governance, Northbridge Institute of Advanced Computing (NIAC). This handbook is binding on all researchers, graduate assistants, and affiliated collaborators working within the Halvorsen AI Systems Laboratory (HASL) and any NIAC-sponsored artificial intelligence research project, regardless of funding source.

1. Mission and Scope

The Halvorsen AI Systems Laboratory exists to conduct reproducible, ethically-governed artificial intelligence research across three focus tracks: Applied Machine Learning, Human-Centered AI, and Computational Trust Systems. This handbook governs every research project registered under any of these three tracks.

Any project not registered with the Office of Research Governance under a valid Project Registration Number (format: NIAC-AI-YYYY-####) is not authorized to use laboratory compute allocations, laboratory datasets, or the HASL GPU cluster.

2. Laboratory Access Rules

Access to the Halvorsen AI Systems Laboratory is tiered into three badge levels: Observer, Associate, and Principal.

Observer badges permit laboratory entry only between 08:00 and 20:00 and do not permit unsupervised use of GPU workstations. Associate badges permit 24-hour access but require a logged safety briefing renewed every 180 days. Principal badges are issued only to project leads and permit access to the Restricted Compute Wing, where GPU nodes reserved for sensitive datasets are housed.

No more than two Observer-badge researchers may occupy the Restricted Compute Wing at any time, and only when accompanied by an Associate or Principal badge holder. Tailgating through the badge-controlled door is treated as a Category 2 security incident under Section 12.

3. Dataset Approval Process

Every AI research project at Northbridge Institute must complete a three-stage dataset review before experimentation may begin. Skipping any stage, even for datasets previously approved for a different project, constitutes a compliance violation.

Stage 1 - Provenance Review. The researcher submits Form NIAC-DR-1 documenting the dataset's origin, collection method, and licensing terms. Stage 1 must be completed within 5 working days of submission by the Data Governance Officer. Datasets scraped from public web sources without a documented collection methodology are automatically rejected at Stage 1.

Stage 2 - Privacy and Bias Screening. The Research Ethics and Data Board (REDB) screens the dataset for personally identifiable information and conducts a demographic representation audit. Stage 2 requires a minimum of 10 working days and cannot be expedited, even for time-sensitive grant deadlines. Any dataset containing unredacted personal data is escalated to a full REDB hearing, which adds a further 15 working days.

Stage 3 - Technical Sign-Off. The Laboratory Director confirms the dataset is compatible with HASL storage encryption requirements (Section 12) and assigns a Dataset Handling Classification of Green, Amber, or Red. Stage 3 requires 3 working days. Red-classified datasets may only be processed on air-gapped Restricted Compute Wing nodes.

The full three-stage review therefore takes a minimum of 18 working days for a Green-classified dataset with no REDB escalation, and researchers are strongly advised to submit Form NIAC-DR-1 at least 25 working days before any planned experimentation date.

4. Experiment Requirements

All experiments must be pre-registered using the HASL Experiment Ledger before any training run consuming more than 4 GPU-hours may begin. The pre-registration must specify the hypothesis, evaluation metric, and a stopping criterion.

Training runs exceeding 200 GPU-hours require co-signature from a Principal badge holder and must include a checkpoint saved no less frequently than every 6 hours to the HASL Cold Storage Array. Failure to checkpoint at this interval is grounds for the run being terminated by laboratory operations staff without warning.

Any experiment involving generative model outputs shown to human evaluators must first pass a Content Safety Pre-Screen conducted by the on-call laboratory safety officer, valid for 30 days from the date of screening.

5. Research Ethics

All projects involving human subjects, human-generated data, or human-evaluated model outputs require approval from the Northbridge Institute Human Subjects Board (HSB) in addition to REDB dataset approval. HSB approval is a separate process from the Section 3 dataset review and typically requires 20 working days.

Deceptive study designs (where participants are not told the true purpose of the study) require a HSB Category B approval, which adds a mandatory 10-day public comment period before the study may proceed.

Researchers must disclose any funding source with a commercial interest in the research outcome in Section 7 of the HSB application. Nondisclosure is treated as a Research Integrity violation under Section 9.

6. Publication Requirements

Any publication, preprint, or public presentation referencing HASL compute resources, HASL datasets, or the Halvorsen AI Systems Laboratory name must be submitted to the Office of Research Governance for a pre-publication compliance check no less than 10 working days before the submission deadline of the target venue.

The compliance check verifies that no Red-classified dataset details are disclosed, that all co-authors hold a current NIAC affiliation or an approved external-collaborator waiver, and that Project Registration Numbers are cited in the acknowledgments section.

Publications resulting from projects that skipped any stage of the Section 3 dataset review will be withdrawn from institutional sponsorship, and the Laboratory Director must notify the venue of the compliance failure within 5 working days of discovery.

7. Review Procedures

Internal project reviews occur at two checkpoints: the Midpoint Review (at 50% of the approved project timeline) and the Closeout Review (within 15 working days of project completion).

The Midpoint Review Committee consists of the Laboratory Director, one REDB representative, and one researcher from a different research track than the project under review. A project failing its Midpoint Review may be granted a single 30-day remediation window before compute allocations are suspended.

The Closeout Review requires submission of a Reproducibility Package containing the exact dataset version hash, environment specification, and random seeds used for all reported results. Projects that cannot produce a Reproducibility Package within 15 working days of a Closeout Review request are marked Non-Reproducible in the institutional research registry.

8. Data Retention Policy

Raw datasets classified Green under Section 3 must be retained for a minimum of 3 years following project closeout. Amber-classified datasets must be retained for 5 years. Red-classified datasets must be retained for 7 years and stored exclusively on the HASL Cold Storage Array with AES-256 encryption at rest.

Trained model checkpoints must be retained for the same duration as the dataset classification used to train them. Intermediate checkpoints not selected for the Reproducibility Package may be deleted after the Closeout Review, provided the final checkpoint is retained.

Upon expiry of the applicable retention period, the Data Governance Officer must certify secure deletion using the NIAC-approved cryptographic erasure procedure and record the deletion in the institutional research registry within 10 working days.

9. Research Team Responsibilities

Every registered project must designate exactly one Principal Investigator, who bears sole responsibility for Section 3 compliance and Section 12 security compliance, and may delegate day-to-day supervision to an Associate-badge co-investigator but not the underlying accountability.

Graduate research assistants may not submit Form NIAC-DR-1 on their own authority; it must be co-signed by their supervising Principal Investigator. A Principal Investigator may supervise no more than 6 active projects concurrently.

Any change of Principal Investigator mid-project (including due to departure from the Institute) must be reported to the Office of Research Governance within 5 working days, and the incoming Principal Investigator must re-affirm Section 3 and Section 5 compliance in writing.

10. Submission Deadlines

Annual compute allocation renewal requests are due on the first working day of March each year. Requests submitted after this date are placed in the Overflow Queue and are only granted allocation if capacity remains after all on-time requests are fulfilled.

Grant-funded projects must submit a Progress Report to the Office of Research Governance every 90 days from the project start date, regardless of whether the external funding body has its own separate reporting schedule.

The annual HASL Research Symposium abstract deadline falls 60 days before the symposium date, with a mandatory dataset-classification disclosure field that, if left blank, results in automatic rejection of the abstract.

11. Evaluation Criteria

Projects are scored annually against four weighted criteria: Scientific Rigor (35%), Reproducibility (25%), Compliance History (25%), and Community Impact (15%). A project scoring below 60 out of 100 on this composite scale is placed on a mandatory Improvement Plan for the following review cycle.

Compliance History is calculated by subtracting 8 points per Category 1 security incident and 3 points per Category 2 security incident (see Section 12) accrued in the prior 12 months, with a floor of zero.

Two consecutive Improvement Plan cycles without measurable score improvement results in referral to the Laboratory Director for a formal project continuation review.

12. Security Requirements

All datasets classified Amber or Red under Section 3 must be encrypted at rest using AES-256 and in transit using TLS 1.3 or later. Storing a Red dataset on any device outside the Restricted Compute Wing, including personal laptops or removable media, is a Category 1 security incident.

Category 1 security incidents require immediate reporting to the Laboratory Director within 2 hours of discovery and trigger an automatic suspension of the responsible researcher's badge pending investigation.

Category 2 security incidents (including badge tailgating per Section 2 and failure to renew an Associate safety briefing) require reporting within 24 hours and result in a formal written warning on the first occurrence and a 90-day access downgrade to Observer level on a second occurrence within any 12-month period.

All Restricted Compute Wing workstations must have full-disk encryption enabled and are subject to unannounced compliance audits by the Data Governance Officer no less than twice per calendar year.